Munireach Nannory

PhD Candidate | MS | Machine Learning FEA Engineer | Predictive Simulation

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PROFESSIONAL SUMMARY

Machine learning and FEA focused PhD researcher with ~3 years of experience developing predictive nonlinear Abaqus/CAE simulations and training simulation-driven ML models to accelerate engineering decisions. Built an end-to-end Python and MATLAB automation workflow on a Linux cluster that executed **100,000+ predictive FEA simulations** and saved **~8,000 hours** of human interaction through standardized setup, run-readiness checks, batch execution, and automated post-processing. Developed deep learning methods, including **1D CNN** and **Deep Kernel Learning (DKL)**, for rapid screening of risk drivers and design trade-offs. Strong cross-functional communicator, translating simulation and ML results into decision-ready guidance for design and test planning in fast-paced environments.

AREAS OF EXPERTISE

PROFESSIONAL SKILLS Predictive Nonlinear FEA (contact, large deformation) | Elastic-Plastic Material Modeling | Load Case and BC Definition | Meshing Strategy and Quality | Sensitivity and Parametric Studies | Post-Processing and Interpretation | V&V and Test Correlation | Cross-Functional Collaboration | Technical Documentation

TECHNICAL SKILLS **Machine Learning (Simulation-driven):** 1D CNN | Neural Networks | Deep Kernel Learning (DKL) | Hyperparameter Tuning | Uncertainty-Aware Modeling | Attention Models | Symbolic Regression (PySR)
Modeling and Simulation: Abaqus/CAE | Nonlinear FEA | Contact Modeling | Elastic-Plasticity
Automation and Data: Python | MATLAB | Git/GitHub | Linux (Cluster) | Batch Workflows | Standardized Post-Processing | Dataset Generation

EDUCATION

- **PhD, Civil and Environmental Engineering** | University of Tennessee, Knoxville | May 2024 - Present
  Focus: Finite Element Method, nanoindentation simulation, automation, machine learning, deep learning
  Honors: Student of the Year, University of Tennessee, Knoxville (2025)
  GPA: 3.97
- **MS, Civil and Environmental Engineering** | University of Tennessee, Knoxville | Aug 2023 - May 2025
- **BS, Civil Engineering** | Pañ̃ñāsāstra University of Cambodia | Nov 2016 – Nov 2022
  Honors: Top ranked scholar (Cohorts 50 and 51)
- **BA, Teaching English as a Foreign Language** | Royal University of Phnom Penh | Feb 2017 to Nov 2022

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (NSF funded), University of Tennessee, Knoxville | Knoxville, TN |Aug 2023–Present

- Developed predictive nonlinear Abaqus/CAE models (contact, large deformation, elastic-plasticity) for design-relevant load cases with verification-minded modeling practices.
- Automated 100,000+ Abaqus simulations, saving 8,000 hours of human interaction.
- Trained simulation-driven ML models (neural networks, Deep Kernel Learning) to learn input-to-response mappings for rapid screening of risk drivers and design trade-offs.
- Performed sensitivity studies and interpolation-based analysis to quantify dominant drivers and support decision-ready guidance for design and test planning.
- Communicated results through technical documentation and decision-ready summaries in cross-functional research settings.

Selected Projects

FEM + 1D CNN Substrate Correction for Thin-Film Nanoindentation | Nov 2024–Present

- Generated paired thin-film and bulk curves via FEM sweeps over E, σ_y, and strength/hardening parameters; resampled to a fixed 200-point normalized depth grid with additional input channels.
- Trained a 1D CNN (Huber loss, noise augmentation) to predict bulk response from thin-film indentation curves.
- Achieved <2% validation MAE on held-out simulations; reduced experimental copper thin-film indentation error by ~60–80% versus uncorrected measurements (with uncertainty bands).
- Enabled improved extraction of bulk-consistent properties including E, σ_y, and ultimate strength after correction.

Substrate-Effect Sensitivity and Regime Mapping (FEM Automation + Data-Driven Guidance) | Jul 2024–Present

- Automated high-throughput simulation campaigns to produce response maps, regime boundaries, and depth-binned error metrics for reliable decision-making.

- Quantified sensitivity trends to reduce trial-and-error and support simulation-driven planning.
- Standardized mesh quality and modeling assumptions across batches and documented recommended meshing practices to balance accuracy and compute cost.

Spherical Tip Geometry Integration for Shallow Indentation | Nov 2023–2025

- Incorporated microscopy-based spherical tip metrology into FEM by using measured tip profiles instead of ideal radii.
- Reduced experiment-to-simulation mismatch from ~20% to <5%, improving shallow-depth agreement and enabling higher-fidelity property extraction.

ACADEMIC ACHIEVEMENTS & RECOGNITIONS

- **Student of the Year,** IRG 2 Graduate Research Assistant (Research Excellence), University of Tennessee, Knoxville, TN, USA (Jul 2025)
- **Best Poster,** IRG 2 Poster Competition, Advanced Materials & Manufacturing, University of Tennessee, Knoxville, TN, USA (Feb 2026)
- **Top-Ranked Scholar (Cohorts 50 and 51), BS in Civil Engineering (Honors Degree)**, Paññāsāstra University of Cambodia, Phnom Penh, Cambodia (Nov 2022)

CONFERENCE PRESENTATIONS

- **USNCCM 18, Chicago, IL (Jul 20 to 24, 2025):** Munireach Nannory, Vivek Chawla, Dayakar Penumadu, Timothy Truster. A novel method for identifying the actual geometry of a spherical nanoindentation tip at shallow depths: Experiments and finite element simulations.
- **ASME IMECE 2024, Portland, OR (Nov 17 to 21, 2024):** Munireach Nannory, Vivek Chawla, Dayakar Penumadu, Timothy Truster. FE nanoindentation simulation for characterization at material extremes. No. IMECE2024 150501.

AWARDS AND FUNDING

- NSF funded PhD Research Assistant, University of Tennessee, Knoxville (Aug 2024 to Present)
- NSF funded MS Research Assistant, University of Tennessee, Knoxville (Aug 2023 to May 2025)

UNIVERSITY SEMINARS AND PRESENTATIONS

Finite Element Overview and Literature Review for Characterization of Materials at Extremes, UTK-MRSEC Center for Advanced Materials and Manufacturing, University of Tennessee, Knoxville, TN (2023)

- **Framework for Incorporating Measured Spherical Indenter Geometry in Abaqus/CAE: 2D vs 3D FEM Compared with Experiments**, UTK-MRSEC Center for Advanced Materials and Manufacturing, University of Tennessee, Knoxville, TN (2024)
- **Advancements in Finite Element Simulation for Nanoindentation: A Novel Method for Identifying Actual Indenter Geometry at Shallow Penetration Depths**, UTK-MRSEC Center for Advanced Materials and Manufacturing, University of Tennessee, Knoxville, TN (2025)
- **Substrate Correction of Nanoindentation Depth–Load Response: FEM and Machine Learning Prediction of Bulk/Film Response from Thin-Film Nanoindentation**, University of Tennessee, Knoxville, TN (2025)

Poster Presentations (Selected)

- **Best Poster (IRG2),** Learning Substrate-Effect Regimes in Thin-Film Indentation: Depth-Normalized Metrics, Sensitivity Analysis, and Active Experiment Design, Advanced Materials & Manufacturing, UTK (2026)
- Multiscale 2D/3D Finite Element Simulation for Nanoindentation: Mesh, Friction, and Surrogate/ML Modeling, Advanced Materials & Manufacturing, UTK (2025)
- FE Nanoindentation Simulations for Characterization at Material Extremes, Advanced Materials & Manufacturing, UTK (2023)

PUBLICATIONS AND MANUSCRIPTS

Selected Publications and Manuscripts *(* lead author)*

- **Munireach Nannory*** et al., "A FEM and Machine Learning Framework for Substrate-Effect Correction in Thin-Film Nanoindentation." **Full draft complete.** *(1D CNN; <2% MAE; ~60–80% experimental error reduction.)*
- **Munireach Nannory*** et al., "Learning Substrate-Effect Regimes in Thin-Film Indentation: Depth-Normalized Metrics, Sensitivity Analysis, and Active Experiment Design." **In preparation.** *(112,200 simulations; regime maps; inverse planner for target error tolerances.)*
- Munireach Nannory et al., "Role of Spherical Tip Geometry in Instrumented Nanoindentation at Shallow Depths." **Under review.**
- **Munireach Nannory*** et al., "Towards Reliable Plastic Property Extraction from Shallow Indentation Data." **Submission in progress**